SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	30 June2026
Team ID	
Project Name	Rising Waters: Smart AI Flood Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (Meteorologists or Disaster Management Officers) can securely log in to the Flood Prediction System using valid credentials before accessing prediction services.	High
2	Authorization levels	Different users are assigned appropriate access levels. Administrators can manage the system, while authorized users can enter weather data, view predictions, and generate reports.	High

3	External interfaces	*The system accepts weather and rainfall data from users and can be integrated with weather APIs or IBM Cloud services for real-time forecasting and deployment.*	Medium
		services the system must connect to]	
4	Transactions processing	The system shall receive weather and rainfall data from users, preprocess the input, execute the trained machine learning model, and return flood prediction results in real time.	High
5	Reporting	The system shall display flood prediction results, prediction probability, and maintain prediction history for future analysis and decision-making.]	Medium
6	Business rules, etc.	The system shall validate all input values before prediction, use the trained XGBoost model for classification, and classify results as **Flood Likely** or **No Flood** based on the prediction threshold.	High
7	Compliance to laws or regulations	The system shall ensure secure handling of user data, follow data privacy guidelines, and comply with organizational and environmental data management policies where applicable.	Medium

8	*Other*	The system shall provide a user-friendly web interface, support future integration with real-time weather APIs, and allow deployment on IBM Cloud for remote accessibility.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	The system should process user input and generate flood predictions quickly with minimal delay.	*Prediction response time ≤ **3 seconds***
2	Scalability	The application should support multiple users simultaneously and allow future deployment on IBM Cloud without performance degradation.	*Supports at least **100 concurrent users** with stable performance.*
3	Security & Data Privacy	User credentials and weather data should be securely stored and protected from unauthorized access using secure authentication and encrypted communication.	***HTTPS enabled**, secure login, and encrypted data storage.)*
4	Reliability & Availability	The system should consistently provide accurate predictions and remain available during disaster situations.	***99% uptime** with prediction accuracy of **96.55%** or higher.*
5	Usability & Accessibility	The web interface should be simple, responsive, and easy to use for meteorologists and disaster management officials on desktop and mobile devices..	*User-friendly interface with responsive design and compatibility across major browsers.*

6	*Other*	The system should allow easy updates to the machine learning model, dataset, and application features, and support deployment on different operating systems and IBM Cloud.	Modular codebase, easy model replacement, and compatibility with **Windows, Linux, macOS, and IBM Cloud**.